
Hadamard-Pratt Arithmetic

A Coordinatewise Product on Pratt Coordinates,
a Semiring on $\mathbb{N}$, and the Spectrum of Its Unitization

Orges Leka
with assistance from ChatGPT 5.6

Draft – August 4, 2026

Abstract. The Pratt-coordinate map sends multiplication of positive integers to addition of finitely supported prime-indexed vectors. We show that its image has an additional and less obvious closure property: the coordinatewise, or Hadamard, product of two natural Pratt vectors is again the Pratt vector of a natural number. This produces a commutative semiring without multiplicative identity on the natural numbers, in which ordinary multiplication plays the role of addition and a new operation $\star_P$ plays the role of multiplication. We give an explicit formula for the ordinary prime exponents of $a \star_P b$, a nonnegative decomposition proving integrality, algorithms and numerical examples, and several structural consequences. The Grothendieck completion is the nonunital coordinate ring $\mathbb{Z}^{(P)}$ with pointwise multiplication, equivalently a ring structure on $\mathbb{Q}_{>0}$. Its canonical unitization is the ring of eventually constant integer sequences. We classify the prime spectrum of this unitization and explain why the bare coordinate ring does not by itself remember the Pratt ancestry relation $q \mid p - 1$: that information remains in the natural Pratt cone and in the Pratt boundary operator.

This note builds on the numerical Pratt-coordinate theory in [1]. No claim of priority for the Hadamard construction is made here; the purpose is to record the structure and its proofs in a reusable form.

Contents

1	Introduction	2
2	Pratt coordinates and the boundary operator	2
2.1	Numerical Pratt coordinates	3
2.2	The transition operator	3
2.3	The natural Pratt cone	4
3	Hadamard closure	4
3.1	A general cone lemma	4
3.2	Closure of natural Pratt vectors	5
4	The induced arithmetic on natural numbers	6
4.1	Explicit prime exponents	6
4.2	Semiring laws	6
4.3	The operation as overlap of Pratt forests	8
5	Numerical examples	8
5.1	Small Pratt vectors	8
5.2	A small multiplication table	9
5.3	A direct algorithm	10
6	The rational ring completion	10
6.1	The Grothendieck group	10
6.2	A ring structure on positive rationals	11
7	Unitization and its prime spectrum	11
7.1	What the spectrum does and does not remember	13
8	Further directions	13
9	Conclusion	14

1 Introduction

For a positive integer n , its ordinary prime-exponent vector

$$v(n) = (v_p(n))_{p \in \mathcal{P}}$$

linearizes multiplication:

$$v(ab) = v(a) + v(b).$$

Pratt coordinates refine this vector by recursively recording the factorizations of $q - 1$ below every prime node q . If

$$M(n) = (m_p(n))_{p \in \mathcal{P}}$$

denotes the full Pratt vector, then one still has

$$M(ab) = M(a) + M(b),$$

but $M(n)$ retains the complete multiplicity profile of all prime labels occurring in the Pratt forest of n .

The question studied here is elementary to state:

If two Pratt vectors are multiplied coordinate by coordinate, is the result again the Pratt vector of a natural number?

For arbitrary nonnegative finite vectors the answer is of course no: not every vector satisfies the recursive consistency conditions of a Pratt forest. The main result is that the answer becomes *yes* when both input vectors already come from natural numbers.

Writing

$$(x \circ y)_p = x_p y_p$$

for the Hadamard product, we prove

$$M(a) \circ M(b) = M(c)$$

for a unique $c \in \mathbb{N}$. We denote this integer by

$$c = a \star_P b.$$

Thus the natural numbers carry two compatible operations:

$$a \oplus_P b := ab, \quad a \otimes_P b := a \star_P b.$$

The first is ordinary multiplication; the second is transported coordinatewise multiplication. These operations form a commutative semiring without a multiplicative identity.

The closure theorem is not a formal consequence of coordinatewise nonnegativity. Its proof uses the Pratt boundary inequalities and the integrality of the transition multiplicities $v_q(p - 1)$. A useful general principle emerges: cones of the form $x \geq Bx$ for a nonnegative integer matrix B are closed under Hadamard products.

2 Pratt coordinates and the boundary operator

Throughout, $\mathcal{P}$ is the set of primes, $\mathbb{N} = \{1, 2, 3, \dots\}$, and a superscript $(\mathcal{P})$ denotes finite support.

2.1 Numerical Pratt coordinates

Definition 2.1 (Pratt exponents). For $p \in \mathcal{P}$, the function $m_p : \mathbb{N} \rightarrow \mathbb{N}_0$ is determined by

$$m_p(1) = 0, \quad m_p(ab) = m_p(a) + m_p(b),$$

and, for a prime q ,

$$m_p(q) = \mathbf{1}_{p=q} + m_p(q-1).$$

The *Pratt vector* of n is

$$M(n) := (m_p(n))_{p \in \mathcal{P}} \in \mathbb{N}_0^{(\mathcal{P})}.$$

The vector has finite support because every edge of a Pratt tree leads from a prime q to prime divisors of $q-1$, hence to smaller labels. The numerical reconstruction formula is

$$n = \prod_{p \in \mathcal{P}} \left(\frac{p}{p-1} \right)^{m_p(n)}. \quad (1)$$

Only finitely many factors differ from 1.

2.2 The transition operator

Define the integer matrix

$$B_{q,p} := v_q(p-1), \quad p, q \in \mathcal{P}.$$

Thus $B_{q,p}$ counts the number of children labelled q immediately below a node labelled p . Since $q \mid p-1$ implies $q < p$, the matrix is strictly triangular when the primes are ordered increasingly.

For a finite vector $x = (x_p)$ define

$$(Bx)_q := \sum_{p \in \mathcal{P}} v_q(p-1)x_p.$$

The sum is finite. The *Pratt boundary* is

$$\beta(x) := (I - B)x, \quad \beta_q(x) = x_q - \sum_p v_q(p-1)x_p. \quad (2)$$

Combinatorially, x_q counts all q -nodes, while the sum counts those q -nodes that occur as children of higher prime nodes. Their difference counts the q -labelled roots.

Proposition 2.2 (Boundary inversion). *For every $n \in \mathbb{N}$,*

$$\beta(M(n)) = v(n).$$

Equivalently,

$$M(n) = v(n) + BM(n). \quad (3)$$

Proof. In the Pratt forest of n , the total number $m_q(n)$ of nodes labelled q is the sum of two disjoint contributions. First, there are $v_q(n)$ roots labelled q . Second, every node labelled p contributes exactly $v_q(p-1)$ children labelled q . Since there are $m_p(n)$ nodes labelled p , the number of non-root q -nodes is

$$\sum_p m_p(n)v_q(p-1).$$

Therefore

$$m_q(n) = v_q(n) + \sum_p m_p(n)v_q(p-1),$$

which is exactly (3). Rearranging gives the result.

Alternatively, apply v_q to the reconstruction formula (1). □

Because B strictly lowers prime labels, it is locally nilpotent on finite vectors: for each $x \in \mathbb{Z}^{(\mathcal{P})}$, one has $B^k x = 0$ for all sufficiently large k . Consequently

$$(I - B)^{-1}x = (I + B + B^2 + \cdots)x, \quad (4)$$

where the series terminates on every finite vector.

2.3 The natural Pratt cone

Definition 2.3. The *natural Pratt cone* is

$$\mathcal{C}_P := M(\mathbb{N}) \subseteq \mathbb{N}_0^{(\mathcal{P})}.$$

Theorem 2.4 (Intrinsic characterization). *For a finite vector $x \in \mathbb{Z}^{(\mathcal{P})}$, the following are equivalent:*

- (i) $x = M(n)$ for some $n \in \mathbb{N}$;
- (ii) $x \in \mathbb{N}_0^{(\mathcal{P})}$ and $\beta(x) \in \mathbb{N}_0^{(\mathcal{P})}$;
- (iii) there is a vector $u \in \mathbb{N}_0^{(\mathcal{P})}$ such that

$$x = (I - B)^{-1}u.$$

In that case

$$u = \beta(x) = v(n), \quad n = \prod_q q^{u_q}.$$

Proof. The implication (i) $\Rightarrow$ (ii) is Proposition 2.2. The equivalence of (ii) and (iii) follows from the finite inverse (4) and the fact that B has nonnegative integer entries.

For (iii) $\Rightarrow$ (i), set $n = \prod_q q^{u_q}$. Proposition 2.2 gives

$$(I - B)M(n) = v(n) = u = (I - B)x.$$

The operator $I - B$ is injective on finite vectors, so $M(n) = x$. $\square$

Thus $\mathcal{C}_P$ is not the whole nonnegative orthant. It is the recursively constrained cone

$$\mathcal{C}_P = \left\{ x \in \mathbb{N}_0^{(\mathcal{P})} : x_q \geq \sum_p v_q(p-1)x_p \text{ for every } q \right\}. \quad (5)$$

3 Hadamard closure

3.1 A general cone lemma

The central inequality is independent of number theory.

Lemma 3.1 (Hadamard cone lemma). *Let J be an index set and let $(a_j)_{j \in J}$ be finitely supported nonnegative integers. Suppose nonnegative families (x_j) and (y_j) satisfy*

$$X \geq \sum_j a_j x_j, \quad Y \geq \sum_j a_j y_j.$$

Then

$$XY \geq \sum_j a_j x_j y_j.$$

Proof. Put

$$U = \sum_j a_j x_j, \quad V = \sum_j a_j y_j.$$

Then $XY \geq UV$. Moreover,

$$\begin{aligned} UV - \sum_j a_j x_j y_j &= \sum_{j \neq k} a_j a_k x_j y_k + \sum_j (a_j^2 - a_j) x_j y_j \\ &\geq 0, \end{aligned}$$

because $a_j^2 - a_j = a_j(a_j - 1) \geq 0$ for every nonnegative integer a_j . $\square$

The integrality of the coefficients is essential to this short proof: for a real coefficient $0 < a_j < 1$, one has $a_j^2 < a_j$.

3.2 Closure of natural Pratt vectors

Theorem 3.2 (Hadamard closure theorem). *If $x, y \in \mathcal{C}_P$, then their coordinatewise product*

$$z = x \circ y, \quad z_p = x_p y_p,$$

also lies in $\mathcal{C}_P$.

Proof. The vector z is nonnegative and finitely supported. Fix a prime q and put

$$a_p = v_q(p - 1).$$

Since $x, y \in \mathcal{C}_P$, the boundary inequalities (5) give

$$x_q \geq \sum_p a_p x_p, \quad y_q \geq \sum_p a_p y_p.$$

Lemma 3.1 yields

$$x_q y_q \geq \sum_p a_p x_p y_p.$$

Hence

$$\beta_q(z) = x_q y_q - \sum_p v_q(p - 1) x_p y_p \geq 0.$$

This holds for every q , so Theorem 2.4 implies $z \in \mathcal{C}_P$. $\square$

The proof gives more than nonnegativity. Write

$$r_q := \beta_q(x), \quad s_q := \beta_q(y),$$

and

$$X_q := \sum_p a_p x_p, \quad Y_q := \sum_p a_p y_p.$$

Then $x_q = r_q + X_q$ and $y_q = s_q + Y_q$, so

$$\begin{aligned} \beta_q(x \circ y) &= r_q s_q + r_q Y_q + s_q X_q \\ &\quad + \sum_{p \neq t} a_p a_t x_p y_t + \sum_p a_p (a_p - 1) x_p y_p. \end{aligned} \tag{6}$$

Every term on the right is a nonnegative integer. Formula (6) is an explicit positivity certificate for every ordinary prime exponent produced by the new operation.

Corollary 3.3 (Support intersection). *For $a, b \in \mathbb{N}$,*

$$\text{supp } M(a \star_P b) = \text{supp } M(a) \cap \text{supp } M(b).$$

Proof. A coordinate of $M(a) \circ M(b)$ is positive exactly when both corresponding coordinates are positive. $\square$

Thus $a \star_P b$ retains exactly the prime labels shared by the two Pratt forests, while multiplying their label multiplicities.

4 The induced arithmetic on natural numbers

Definition 4.1 (Hadamard-Pratt product). For $a, b \in \mathbb{N}$, define $a \star_P b$ to be the unique natural number satisfying

$$M(a \star_P b) = M(a) \circ M(b). \quad (7)$$

Existence follows from Theorem 3.2; uniqueness follows from the injectivity of Pratt reconstruction.

4.1 Explicit prime exponents

Applying the boundary operator to (7) gives a direct computational formula.

Theorem 4.2 (Exponent formula). *For every prime q ,*

$$v_q(a \star_P b) = m_q(a)m_q(b) - \sum_{p \in \mathcal{P}} v_q(p-1)m_p(a)m_p(b). \quad (8)$$

The right-hand side is always a nonnegative integer.

Proof. Equation (8) is precisely

$$v_q(a \star_P b) = \beta_q(M(a) \circ M(b)).$$

Its nonnegativity is Theorem 3.2, or more explicitly the decomposition (6). $\square$

Consequently

$$a \star_P b = \prod_{q \in \mathcal{P}} q^{m_q(a)m_q(b) - \sum_p v_q(p-1)m_p(a)m_p(b)}. \quad (9)$$

Although the exponents in (9) look like differences, the closure theorem proves that no denominator occurs for natural inputs.

4.2 Semiring laws

Theorem 4.3 (Hadamard-Pratt semiring). *Define on $\mathbb{N}$*

$$a \oplus_P b := ab, \quad a \otimes_P b := a \star_P b.$$

Then $(\mathbb{N}, \oplus_P, \otimes_P)$ is a commutative semiring without multiplicative identity. More explicitly:

- (i) $\oplus_P$ is associative and commutative, with zero element 1;
- (ii) $\otimes_P$ is associative and commutative;
- (iii) 1 is absorbing for $\otimes_P$;

(iv) $\otimes_P$ distributes over $\oplus_P$:

$$a \star_P (bc) = (a \star_P b)(a \star_P c).$$

There is no element of $\mathbb{N}$ serving as a multiplicative identity for $\otimes_P$.

Proof. Under the injective map M , the operation $\oplus_P$ becomes vector addition and $\otimes_P$ becomes coordinatewise multiplication. All stated algebraic laws therefore follow from the corresponding laws in $\mathbb{N}_0^{(P)}$. Since $M(1) = 0$, the ordinary integer 1 is the additive zero and is absorbing for the Hadamard product.

Suppose $e \in \mathbb{N}$ were a $\otimes_P$ -identity. For every prime p , the p -coordinate of $M(p)$ equals 1. Hence

$$M(e)_p M(p)_p = M(p)_p$$

forces $M(e)_p = 1$. This would hold for every prime p , so $M(e)$ would have infinite support, a contradiction. $\square$

A semiring without multiplicative identity is sometimes called a *hemiring*. We retain the more transparent phrase “semiring without identity.”

Corollary 4.4 (Powers and bilinearity). *For $a, b \in \mathbb{N}$ and $r, s \in \mathbb{N}_0$,*

$$a^r \star_P b^s = (a \star_P b)^{rs}.$$

In particular,

$$a \star_P b^s = (a \star_P b)^s.$$

Proof. Since $M(a^r) = rM(a)$ and $M(b^s) = sM(b)$,

$$M(a^r) \circ M(b^s) = rs(M(a) \circ M(b)).$$

The right-hand side is $M((a \star_P b)^{rs})$. $\square$

Proposition 4.5 (Terminal-coordinate extraction). *For every $n \in \mathbb{N}$,*

$$2 \star_P n = 2^{m_2(n)}.$$

Proof. Since $M(2) = e_2$, one has

$$M(2) \circ M(n) = m_2(n)e_2 = M(2^{m_2(n)}).$$

$\square$

Proposition 4.6 (Zero divisors and idempotents). *In the semiring $(\mathbb{N}, \oplus_P, \otimes_P)$:*

- (i) *there are no zero divisors: $a \star_P b = 1$ implies $a = 1$ or $b = 1$;*
- (ii) *the $\star_P$ -idempotents are exactly 1, 2, 3.*

Proof. If $n > 1$, then its Pratt forest contains at least one terminal node labelled 2, so $m_2(n) > 0$. Hence for $a, b > 1$,

$$m_2(a \star_P b) = m_2(a)m_2(b) > 0,$$

and therefore $a \star_P b \neq 1$.

If $n \star_P n = n$, then $m_p(n)^2 = m_p(n)$ for every prime p , so every Pratt coordinate is 0 or 1. In particular, $m_2(n) \leq 1$. If n is composite, write $n = ab$ with $a, b > 1$; complete additivity gives

$$m_2(n) = m_2(a) + m_2(b) \geq 2.$$

If $n = p$ is prime and $p > 3$, then $p - 1 = 2k$ with $k \geq 2$, whence

$$m_2(p) = m_2(p - 1) = 1 + m_2(k) \geq 2.$$

Thus only $n = 1, 2, 3$ are possible, and direct inspection shows that all three are idempotent. $\square$

4.3 The operation as overlap of Pratt forests

If a Pratt forest is represented only by its multiplicity vector, then ordinary multiplication concatenates forests:

$$M(ab) = M(a) + M(b).$$

The new product instead takes the overlap of label supports and multiplies multiplicities:

$$m_p(a \star_P b) = m_p(a)m_p(b).$$

Thus the two operations have complementary interpretations:

Operation on integers	Operation on Pratt vectors	Forest interpretation
ab	$M(a) + M(b)$	disjoint union of forests
$a \star_P b$	$M(a) \circ M(b)$	multiplicative overlap of labels

5 Numerical examples

We write e_p for the vector having coordinate 1 at p and 0 elsewhere.

5.1 Small Pratt vectors

n	$M(n)$
1	0
2	e_2
3	$e_2 + e_3$
4	$2e_2$
5	$2e_2 + e_5$
6	$2e_2 + e_3$
7	$2e_2 + e_3 + e_7$
10	$3e_2 + e_5$
11	$3e_2 + e_5 + e_{11}$
12	$3e_2 + e_3$
18	$3e_2 + 2e_3$

Example 5.1 ($3 \star_P 5 = 4$). One has

$$M(3) = e_2 + e_3, \quad M(5) = 2e_2 + e_5.$$

Therefore

$$M(3) \circ M(5) = 2e_2 = M(4),$$

so

$$\boxed{3 \star_P 5 = 4.}$$

Only the terminal label 2 is shared by the two Pratt trees.

Example 5.2 ($5 \star_P 5 = 20$). Here

$$M(5) \circ M(5) = 4e_2 + e_5.$$

The ordinary boundary exponents are

$$\beta_2 = 4 - v_2(5 - 1) = 4 - 2 = 2, \quad \beta_5 = 1.$$

Hence

$$\boxed{5 \star_P 5 = 2^2 \cdot 5 = 20.}$$

Example 5.3 ($7 \star_P 7 = 28$). Since

$$M(7) = 2e_2 + e_3 + e_7,$$

the Hadamard square is

$$4e_2 + e_3 + e_7.$$

Its boundary is

$$\begin{aligned}\beta_2 &= 4 - v_2(3 - 1) - v_2(7 - 1) = 4 - 1 - 1 = 2, \\ \beta_3 &= 1 - v_3(7 - 1) = 1 - 1 = 0, \\ \beta_7 &= 1.\end{aligned}$$

Thus

$$\boxed{7 \star_P 7 = 2^2 \cdot 7 = 28.}$$

The internal 3-node cancels at the boundary, exactly as it should.

Example 5.4 ($7 \star_P 11 = 64$). The vectors are

$$M(7) = 2e_2 + e_3 + e_7, \quad M(11) = 3e_2 + e_5 + e_{11}.$$

Their only common Pratt label is 2, so

$$M(7) \circ M(11) = 6e_2 = M(2^6).$$

Therefore

$$\boxed{7 \star_P 11 = 64.}$$

Example 5.5 ($12 \star_P 18 = 1152$). Using

$$M(12) = 3e_2 + e_3, \quad M(18) = 3e_2 + 2e_3,$$

we obtain

$$M(12) \circ M(18) = 9e_2 + 2e_3.$$

The boundary exponents are

$$\beta_2 = 9 - 2 = 7, \quad \beta_3 = 2.$$

Hence

$$\boxed{12 \star_P 18 = 2^7 \cdot 3^2 = 1152.}$$

5.2 A small multiplication table

The following table records $p \star_P q$ for the first six primes.

$\star_P$	2	3	5	7	11	13
2	2	2	4	4	8	8
3	2	3	4	6	8	12
5	4	4	20	16	80	64
7	4	6	16	28	64	96
11	8	8	80	64	704	512
13	8	12	64	96	512	832

The table illustrates several features. Values can be much larger than either input; pairs with no shared nonterminal Pratt label produce powers of 2; and the operation is visibly symmetric but very different from ordinary multiplication.

5.3 A direct algorithm

Given $a, b \in \mathbb{N}$:

1. Compute the finite Pratt vectors $x = M(a)$ and $y = M(b)$.
2. Form $z_p = x_p y_p$.
3. For every relevant prime q , compute

$$e_q = z_q - \sum_p v_q(p-1)z_p.$$

4. Return

$$a \star_P b = \prod_q q^{e_q}.$$

Theorem 3.2 guarantees $e_q \geq 0$, so the algorithm never produces a denominator for natural inputs.

6 The rational ring completion

6.1 The Grothendieck group

The additive monoid $(\mathcal{C}_P, +)$ is isomorphic to $(\mathbb{N}, \cdot)$. Its Grothendieck group is therefore $(\mathbb{Q}_{>0}, \cdot)$. On coordinates this completion is all of $\mathbb{Z}^{(\mathcal{P})}$.

Proposition 6.1. *The additive subgroup of $\mathbb{Z}^{(\mathcal{P})}$ generated by $\mathcal{C}_P$ is the whole group $\mathbb{Z}^{(\mathcal{P})}$.*

Proof. For every prime p , the defining recursion gives

$$M(p) = e_p + M(p-1).$$

Therefore

$$e_p = M(p) - M(p-1). \tag{10}$$

Every finite integer vector is an integer linear combination of the basis vectors e_p , hence lies in the group generated by natural Pratt vectors. $\square$

The Pratt map extends uniquely to

$$\widetilde{M} : \mathbb{Q}_{>0} \longrightarrow \mathbb{Z}^{(\mathcal{P})}, \quad \widetilde{M}(a/b) = M(a) - M(b).$$

It is an isomorphism of abelian groups from ordinary multiplication on $\mathbb{Q}_{>0}$ to coordinate addition on $\mathbb{Z}^{(\mathcal{P})}$.

The inverse is the reconstruction map

$$\mathcal{R}(x) = \prod_p \left(\frac{p}{p-1} \right)^{x_p}, \quad x \in \mathbb{Z}^{(\mathcal{P})}. \tag{11}$$

This map is injective. Indeed, the ordinary valuation vector of $\mathcal{R}(x)$ is $\beta(x)$; if $\mathcal{R}(x) = 1$, then $\beta(x) = 0$, and the invertibility of $I - B$ forces $x = 0$.

6.2 A ring structure on positive rationals

Coordinatewise multiplication makes

$$R_P := \mathbb{Z}^{(P)}$$

a commutative ring without multiplicative identity. Transporting this structure through $\mathcal{R}$ gives two operations on $\mathbb{Q}_{>0}$:

$$r \oplus_P s := rs, \quad r \otimes_P s := \mathcal{R}(\widetilde{M}(r) \circ \widetilde{M}(s)).$$

Then $(\mathbb{Q}_{>0}, \oplus_P, \otimes_P)$ is a commutative ring without identity. In this ring:

- the additive zero is the ordinary number 1;
- the additive inverse of r is the ordinary reciprocal r^{-1} ;
- the product $\otimes_P$ restricts to $\star_P$ on natural numbers;
- for arbitrary rational inputs, the result need not be an integer.

Example 6.2 (Orthogonal rational idempotents). Equation (10) and reconstruction give

$$\mathcal{R}(e_p) = \frac{p}{p-1}.$$

Because

$$e_p \circ e_p = e_p, \quad e_p \circ e_q = 0 \quad (p \neq q),$$

one obtains

$$\frac{p}{p-1} \otimes_P \frac{p}{p-1} = \frac{p}{p-1},$$

and

$$\frac{p}{p-1} \otimes_P \frac{q}{q-1} = 1 \quad (p \neq q).$$

For example,

$$\frac{3}{2} \otimes_P \frac{3}{2} = \frac{3}{2}, \quad \frac{3}{2} \otimes_P \frac{5}{4} = 1.$$

Thus the coordinate directions become mutually orthogonal idempotents in the rational completion.

Remark 6.3. This ring is highly non-domain-like: it has infinitely many nonzero orthogonal idempotents. Its multiplication is diagonal and contains no interaction between distinct prime coordinates. The nontrivial arithmetic closure of $\mathcal{C}_P$ comes from the shape of the positive cone, not from the ambient ring alone.

7 Unitization and its prime spectrum

Ordinary scheme theory usually works with rings having a multiplicative identity. There are two obvious ways to add one. The full product $\mathbb{Z}^P$ has the all-ones vector, but it introduces arbitrary infinite sequences and an enormous ultrafilter spectrum. A smaller canonical choice is the unitization of R_P .

Definition 7.1 (Canonical unitization). Let

$$\widehat{R}_P := \mathbb{Z}\mathbf{1} \oplus R_P.$$

Equivalently,

$$\widehat{R}_P = \left\{ (a_p)_{p \in \mathcal{P}} \in \mathbb{Z}^{\mathcal{P}} : (a_p) \text{ is eventually constant} \right\},$$

with pointwise addition and multiplication. Its identity is the constant sequence $\mathbf{1} = (1, 1, 1, \dots)$.

Let

$$X := \mathcal{P} \cup \{\infty\}$$

be the one-point compactification of the discrete set of primes. An eventually constant sequence is the same thing as a continuous function $X \rightarrow \mathbb{Z}$ when $\mathbb{Z}$ is given the discrete topology. For $x \in X$, write

$$\text{ev}_x : \widehat{R}_P \rightarrow \mathbb{Z}$$

for evaluation at x , where evaluation at ∞ means the eventual constant value.

Theorem 7.2 (Prime spectrum of the unitization). *Every prime ideal of $\widehat{R}_P$ has the form*

$$\mathfrak{P}_{x,\mathfrak{q}} := \text{ev}_x^{-1}(\mathfrak{q}), \quad x \in X, \quad \mathfrak{q} \in \text{Spec } \mathbb{Z}.$$

Moreover, the resulting bijection is a homeomorphism

$$\boxed{\text{Spec}(\widehat{R}_P) \cong X \times \text{Spec } \mathbb{Z}.}$$

Proof. Let $\mathfrak{P}$ be a prime ideal of $\widehat{R}_P$. The vectors e_p are pairwise orthogonal idempotents:

$$e_p e_q = 0 \quad (p \neq q).$$

Hence at most one e_p can lie outside $\mathfrak{P}$; otherwise primality would be contradicted by a product of two elements outside $\mathfrak{P}$ equal to zero.

Suppose first that $e_r \notin \mathfrak{P}$ for one prime r . From

$$e_r(1 - e_r) = 0$$

and primality, one gets $1 - e_r \in \mathfrak{P}$. Also $e_q \in \mathfrak{P}$ for every $q \neq r$. These elements generate $\ker(\text{ev}_r)$, so

$$\ker(\text{ev}_r) \subseteq \mathfrak{P}.$$

The quotient by this kernel is $\mathbb{Z}$, and therefore

$$\mathfrak{P} = \text{ev}_r^{-1}(\mathfrak{q})$$

for a unique $\mathfrak{q} \in \text{Spec } \mathbb{Z}$.

Suppose instead that every e_p belongs to $\mathfrak{P}$. Then the finite-support ideal R_P is contained in $\mathfrak{P}$. Since

$$\widehat{R}_P / R_P \cong \mathbb{Z}$$

via evaluation at ∞ , one has

$$\mathfrak{P} = \text{ev}_\infty^{-1}(\mathfrak{q})$$

for a unique $\mathfrak{q} \in \text{Spec } \mathbb{Z}$.

This proves the bijection. For topology, every $f \in \widehat{R}_P$ has finite image and clopen level sets in X . Under the bijection, the basic open set $D(f)$ becomes

$$\bigcup_{n \in f(X)} f^{-1}(n) \times D_{\mathbb{Z}}(n),$$

which is open in $X \times \text{Spec } \mathbb{Z}$. Conversely, sets of the form $U \times D_{\mathbb{Z}}(n)$ with $U \subseteq X$ clopen arise as basic opens from the function equal to n on U and 0 outside U . Hence the bijection is a homeomorphism. $\square$

7.1 What the spectrum does and does not remember

The classification is explicit, but it reveals a limitation. The abstract coordinate ring

$$R_P = \bigoplus_{p \in \mathcal{P}} \mathbb{Z}e_p$$

depends only on the index set $\mathcal{P}$. It does not use the transition numbers $v_q(p-1)$. Consequently its unitized spectrum remembers a copy of $\text{Spec } \mathbb{Z}$ over every prime coordinate and one additional copy at infinity, but it does not remember the directed Pratt relation

$$q \mid p - 1.$$

The recursive arithmetic is carried by two additional pieces of data:

$$\mathcal{C}_P = (I - B)^{-1} \mathbb{N}_0^{(\mathcal{P})}$$

and

$$\beta = I - B.$$

Thus a more faithful geometric object should retain at least the decorated triple

$$\boxed{(\widehat{R}_P, \mathcal{C}_P, \beta)}$$

or an operator algebra generated by the diagonal coordinate ring together with the transition operator B . The Hadamard closure theorem says that $\mathcal{C}_P$ is a multiplicatively closed positive cone inside this diagonal ring.

8 Further directions

The construction suggests several concrete questions.

1. **Arithmetic classification.** Determine formulas or asymptotics for $a \star_P b$ in special families, for example when a, b are primes, prime powers, Fermat numbers, or numbers with prescribed Pratt support.
2. **Divisibility and order.** Study how $\star_P$ interacts with ordinary divisibility and with the coordinatewise Pratt order

$$a \preceq_P b \iff m_p(a) \leq m_p(b) \text{ for all } p.$$

The operation is automatically monotone for this order.

3. **Operator enrichment.** Adjoin the boundary or transition operator to the diagonal ring and study the algebra generated by multiplication operators and B . Unlike the bare coordinate ring, this operator algebra can remember the Pratt graph.
4. **Polynomial analogue.** The polynomial Pratt-coordinate theory has an analogous boundary operator. One may ask whether its natural polynomial coordinate cone is also closed under Hadamard products. The numerical proof uses the integer multiplicities $v_q(p-1)$ and therefore strongly suggests a corresponding result whenever the polynomial transition multiplicities are nonnegative integers and the image cone admits the same boundary characterization.
5. **Geometry.** The spectrum of the unitized diagonal ring is elementary. A genuinely Pratt-sensitive spectrum would need to incorporate the cone, the transition operator, or the category of Pratt paths rather than only pointwise coordinate multiplication.

9 Conclusion

The Pratt-coordinate embedding does more than linearize ordinary multiplication. Its natural image

$$\mathcal{C}_P = M(\mathbb{N})$$

is unexpectedly closed under coordinatewise multiplication. This produces a second arithmetic operation $\star_P$ on natural numbers, characterized by

$$m_p(a \star_P b) = m_p(a)m_p(b).$$

The ordinary prime exponents of the result are recovered by the Pratt boundary and possess the explicit nonnegative formula (8). Hence the integrality of $a \star_P b$ is a theorem, not an experimental coincidence.

After group completion, the construction becomes the diagonal nonunital ring $\mathbb{Z}^{(\mathcal{P})}$, or equivalently a ring structure on $\mathbb{Q}_{>0}$ in which ordinary multiplication is addition. The canonical unitization has a completely describable spectrum. That spectrum is useful as a baseline, but it also shows where the deepest Pratt information lives: not in the diagonal ring alone, but in the distinguished natural cone and in the boundary operator encoding the recursive relation $q \mid p - 1$.

From this viewpoint, natural numbers are not merely points in a coordinate space. They form a highly constrained positive cone that is simultaneously closed under vector addition and Hadamard multiplication. This double closure is what makes the Hadamard-Pratt arithmetic both rigorous and unexpectedly enjoyable.

References

- [1] O. Leka, with assistance from ChatGPT 5.6, *Pratt Coordinates for Natural Numbers and Monic Rational Polynomials: Reconstruction, Hilbert-Space Embeddings, and a Pratt-Coordinate Form of Mason–Stothers*, Version 0.2, August 3, 2026.